# WEB DEVELOPMENT 1

**Learning Groups: P107, P108, E101, P102**

**Lecturer: Mr. Maina Samuel**

**week 3** **learning topics**

- **3.0 HTML table**
- **3.1 HTML blocks**
- **3.2 HTML id and classes**
- **3.3 HTML layout**

## 3.0 HTML tables

A table is **a grid of rows and columns**, the intersections of which form cells. Each cell is a distinct area, into which you can place text, graphics, or even other tables.

HTML handles tables very well, and you can use them to organize and present complex data to your site visitors. For example, you could display your store's inventory in a table.

### Basic Table Structure

Tables are defined with the **<table>** tag.

Tables are divided into **table rows** with the **<tr>** tag.

Table rows are divided into **table data** with the **<td>** tag.

A table row can also be divided into **table headings** with the **<th>** tag.

Edit This Code: See Result »

```
<!DOCTYPE html>
<html>
<head>
        <title>html table</title>
</head>
<body>
        <table>
                <tr>
                        <td>15</td>
                        <td>15</td>
                        <td>30</td>
                </tr>
                <tr>
                        <td>45</td>
                        <td>60</td>
                        <td>45</td>
                </tr>
                <tr>
                        <td>60</td>
                        <td>90</td>
                        <td>90</td>
                </tr>
        </table>

</body>
```

Result:

15 15 30
45 60 45
60 90 90

## Html table Spanning column

chapter-06/spanning-columns.html HTML

```
<table>
  <tr>
    <th></th>
    <th>9am</th>
    <th>10am</th>
    <th>11am</th>
    <th>12am</th>
  </tr>
  <tr>
    <th>Monday</th>
    <td colspan="2">Geography</td>
    <td>Math</td>
    <td>Art</td>
  </tr>
  <tr>
    <th>Tuesday</th>
    <td colspan="3">Gym</td>
    <td>Home Ec</td>
  </tr>
</table>
```

RESULT

| | 9am | 10am | 11am | 12am |
|---|---|---|---|---|
| Monday | Geography | | Math | Art |
| Tuesday | Gym | | | Home Ec |

**Sometimes you may need the entries in a table to stretch across more than one column.**

The **colspan attribute** can be used on a **<th>** or **<td>**element and indicates how many columns that cell **should run across (merging cells)**

In the example on the right you can see a **timetable with five columns**; the first column contains the **heading for that row (the day), the remaining four represent one-hour time slots.**

If you look at the table cell that contains the words **'Geography'** you will see that the value of the colspan attribute is 2, which indicates that the cell should run **across two columns.**

In the **third row, 'Gym' runs across three columns.**

## Html table Spanning rows

HTML chapter-06/spanning-rows.html

```
<table>
  <tr>
    <th></th>
    <th>ABC</th>
    <th>BBC</th>
    <th>CNN</th>
  </tr>
  <tr>
    <th>6pm - 7pm</th>
    <td rowspan="2">Movie</td>
    <td>Comedy</td>
    <td>News</td>
  </tr>
  <tr>
    <th>7pm - 8pm</th>
    <td>Sport</td>
    <td>Current Affairs</td>
  </tr>
</table>
```

RESULT

| | ABC | BBC | CNN |
|---|---|---|---|
| 6pm - 7pm | Movie | Comedy | News |
| 7pm - 8pm | | Sport | Current Affairs |

**You may also need entries in a table to stretch down across more than one row.**

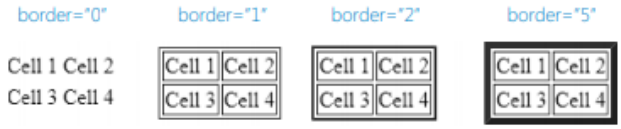
- **The row span attribute** can be used on a<td> or<td> element to indicate **how many rows a cell should span down the table.**
- In the example on the left you can see that **ABC Tv** is showing a movie from 6pm – 8pm, whereas the **BBC and CNN** channels **are both showing two programs** during this time period **(each of which lasts one hour).**
- If you look at the last element, it only contains three elements even though there **are four columns in the result below.** This is because the movie in the**<tr>**

## Applying Borders by Using Attributes

By default, **a table has no border**. To add a one-pixel border around both the table as a whole and around each individual cell, you can add this attribute to the **<table>** tag

**<table border =”1”>**

As shown in the following examples, increasing the number increases the width of the outer border around the whole table, but not the inner borders:



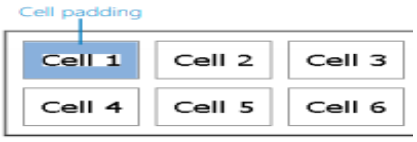

**Difference between Border and padding**

- **Border** –Border is Space outside padding. has no space inside cell?
- **Padding**- space inside the border

## Changing Cell Padding, Spacing, and Alignment

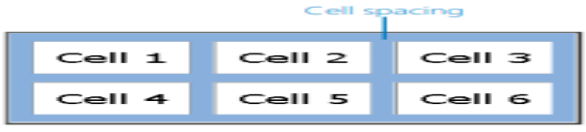
Cell padding, cell spacing, and cell alignment are ***three*** different ways you can control how cell content appears on a page.

- **Cell Padding** -refers to the amount of space between an **element’s content** and **its outer edge.** For a table cell, padding refers to space between the cell border and the text or graphic within it.




**padding represented in the shaded region space between border and the text.**

- **Cell Spacing** – refers to the amount of space between the outside of an element and the adjacent element. For a table cell, spacing refers to the space between the border of one cell and the border of the adjacent cell.




- **Alignment** -refers to the placement of the content within its allotted area, either vertically or horizontally. For normal paragraphs (not in a table), alignment refers only to horizontal placement between the margins. For a table

cell, however, there are separate settings for vertical and horizontal alignment.

## Styling HTML table even and odd cell

**Output**

```
<!DOCTYPE>
<html>
<head>
<style>
table, th, td {
    border: 1px solid black;
    border-collapse: collapse;
}
th, td {
    padding: 10px;
}
table#alter tr:nth-child(even) {
    background-color: #eee;
}
table#alter tr:nth-child(odd) {
    background-color: #fff;
}
table#alter th {
    color: white;
    background-color: gray;
}
</style>
</head>
<body>

<table id="alter">
<tr><th>First_Name</th><th>Last_Name</th><th>Marks</th></tr>
<tr><td>Sonoo</td><td>Jaiswal</td><td>60</td></tr>
<tr><td>James</td><td>William</td><td>80</td></tr>
<tr><td>Swati</td><td>Sironi</td><td>82</td></tr>
<tr><td>Chetna</td><td>Singh</td><td>72</td></tr>
</table>
</body>
</html>
```

| First_Name | Last_Name | Marks |
|---|---|---|
| Sonoo | Jaiswal | 60 |
| James | William | 80 |
| Swati | Sironi | 82 |
| Chetna | Singh | 72 |

**What does nth-child mean?**

- **It means select the siblings for styling alternate table row** `<tr>`, **Table alter tr: nth-child(odd)** but only the odd number among all siblings, from the perspective **of their parent element.**

# 3.1 HTML blocks

## HTML Block and Inline Elements

Every HTML element has a default display value depending on what type of element it is. The default display value for most elements is block or inline.

## A block-level Elements using  tag

**A block-level element always starts on a new line** and takes up the full width available (stretches out to the left and right as far as it can).

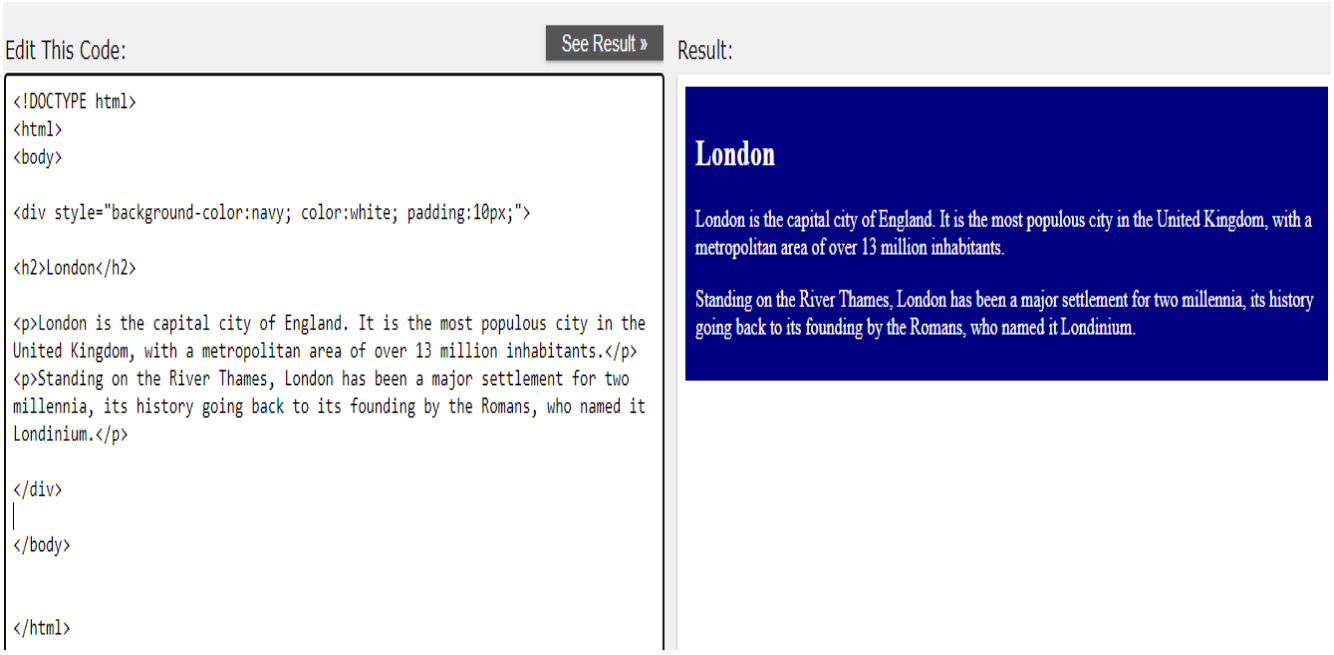
The ** ** element is a block-level element.

**Examples of block-level elements:**

- 
- <h1> - <h6>
- <p>
- <form>

The  element has no required attributes, but **style** and **class** are common.

When used together with CSS, the  element can be used to style blocks of content:



## Inline Elements using  tag

An inline element does not start on a new line and only takes up as much width as necessary.

This is `an inline <span> element inside` a paragraph.

Examples of inline elements:

- 
- <a>
- <img>

The  element is an **inline element** that is often used as a container for some text.




### HTML Grouping Summary Tags

| Tag | Description |
|---|---|
|  | Defines a section in a document (block-level) |
|  | Defines a section in a document (inline) |

## 3.2 HTML Id and classes

**HTML Id Attribute** **NB: * very important subtopic to learn and understand in web development**

The **id attribute** is used to specify the unique ID for an element of the HTML document. **It allocates the unique identifier which is** *used by the* **CSS** *and the* **JavaScript** *for performing certain tasks.*

> ***Note: In the Cascading Style sheet (CSS), we can easily select an element with the specific id by using the # symbol followed by id.***

> ***Note: JavaScript can access an element with the given ID by using the getElementById () method.***

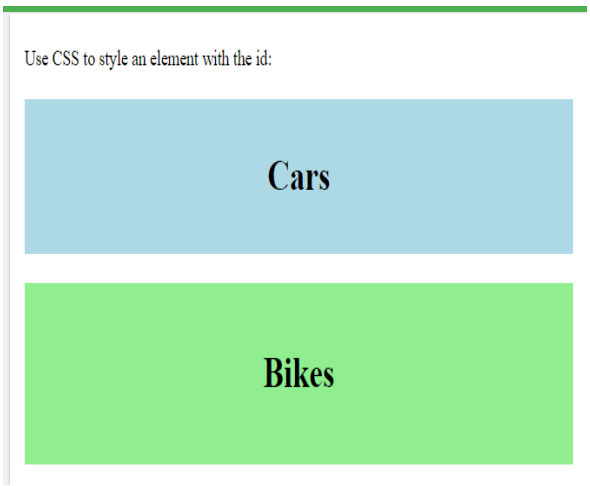
**Syntax:** `<tag id="value">`

**Example illustrating Html Id**

```
<!DOCTYPE html>
<html>
<head>
<title>
Example of Id attribute in CSS
</title>
<style>
#Cars {
padding: 40px;
background-color: lightblue;
color: black;
text-align: center;
}

#Bikes
{
padding: 50px;
background-color: lightGreen;
text-align: center;
}
</style>
</head>
<body>
<p> Use CSS to style an element with the id: </p>
<h1 id="Cars"> Cars </h1>
<h1 id="Bikes"> Bikes </h1>
</body>
</html>
```

**output**




**Points to remember**

**When using id in CSS (cascading style sheet)**

- ID Its detonated with # symbol
- Two different color defining **cars** and **bikes** represents the **uniqueness of id**

## HTML Classes

### Class Attribute in HTML

The HTML class attribute is used to specify a single or multiple class names for an HTML element.

A class attribute can be defined **within <style> tag or in separate file using the (.)** character.

In an HTML document, **we can use the same class attribute name with different elements.**

Example:

```
<head>
  <style>
    .headings{
      color: lightgreen;
      font-family: cursive;
      background-color: black; }
  </style>
</head>
```

Applying class attribute using the **style tag** at Html head section and **using (.) character** to represent class attribute.

**Example 1 using class attribute**. (same color to all headings using class (.) character, to select multiple headings **at once** you control all using Class Element)

Run ▶ Online Code Editor

```
<!DOCTYPE html>
<html>
<head>
	<style>
		.headings{
			color: lightgreen;
			font-family: cursive;
			background-color: black; }
	</style>
</head>
<body>
<h1 class="headings">This is first heading</h1>
<h2 class="headings">This is Second heading</h2>
<h3 class="headings">This is third heading</h3>
<h4 class="headings">This is fourth heading</h4>
</body>
</html>
```

This is first heading

This is Second heading

This is third heading

This is fourth heading

**When should I use class or ID**? – ***id is used for single elements*** *that appear on the page for only once* (e.g. header, footer, menu etc).

Whereas class is used for single or multiple elements that appear on the page **for once or more than once** (e.g paragraph, links, buttons, input boxes)

## Another Example with different class name

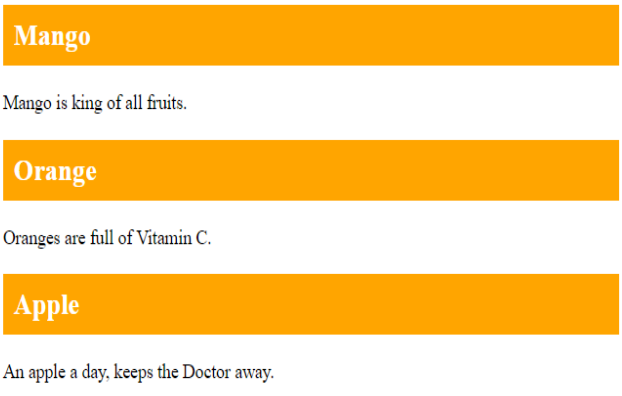
**Let's use a class name "Fruit" with CSS to style all elements.**

```
<style>
.fruit {
    background-color: orange;
    color: white;
    padding: 10px;
}
</style>

<h2 class="fruit">Mango</h2>
<p>Mango is king of all fruits.</p>

<h2 class="fruit">Orange</h2>
<p>Oranges are full of Vitamin C.</p>

<h2 class="fruit">Apple</h2>
<p>An apple a day, keeps the Doctor away.</p>
```




## Class Attribute in JavaScript

You can use JavaScript access elements with a specified class name by using the **`getElementsByClassName`** `()` method

output

```
<!DOCTYPE html>
<html>
<body>

<h2>Class Attribute with JavaScript</h2>
<p>Click the button, to hide all elements with the class name
"fruit", with JavaScript:</p>

<button onclick="myFunction()">Hide elements</button>

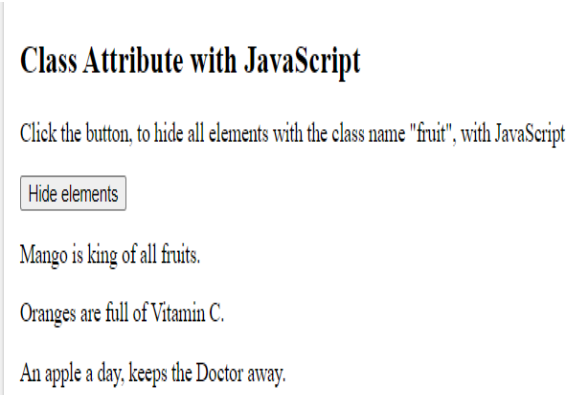

<h2 class="fruit">Mango</h2>
<p>Mango is king of all fruits.</p>

<h2 class="fruit">Orange</h2>
<p>Oranges are full of Vitamin C.</p>

<h2 class="fruit">Apple</h2>
<p>An apple a day, keeps the Doctor away.</p>

<script>
function myFunction() {
  var x = document.getElementsByClassName("fruit");
  for (var i = 0; i < x.length; i++) {
    x[i].style.display = "none";
  }
}
</script>

</body>
</html>
```




- **Using JavaScript to access HTML document we use a method.**
- **GetElementByclassName- <h2 class=" fruit "mango</h2>** declared in Html document to access that in java script we use method called **GetElementByclassName** to select Html class element and its value "fruits"
- **Var x –** variable **NB**: *in JavaScript data type is not declared.*

## 3.3 Html layout

Creating a **website layout** is the activity of positioning the various elements that make a web page in a well-structured manner and give appealing look to the website

Websites often display content in multiple columns (like a magazine or newspaper).

Most **modern websites** and blogs consist of a header, footer, navbar, perhaps another sidebar, and let's not forget, the main content area. Something like this:




- **<header>** - Defines a header for a document or a section
- **<nav>** - Defines a container for navigation links
- **<section>** - Defines a section in a document
- **<article>** - Defines an independent self-contained article
- **<aside/Ads>** - Defines content aside from the content (**like a sidebar)**
- **<footer>** - Defines a footer for a document or a section
- **<details>** - Defines additional details
- **<summary>** - Defines a heading for the <details> element

**Different methods of creating HTML layout**

- HTML Table Based Layout
- HTML Div Based Layout
- Using the HTML5 Structural Elements

**Warning:** Avoid tables and inline styles for creating layouts. Layouts created using tables often rendered very slowly. Tables should only be used to display tabular data.

**Tip:** Better web page layouts can be created by using the DIV elements and CSS. You can change the layout of all the pages of your website by editing just one CSS file. To learn about CSS in detail, please check out our CSS tutorial section.

## HTML Div Based Layout

Using the  elements is the most common method of creating layouts in HTML. The  (stands for division) element is used for marking out a block of content, or set of other elements inside an HTML document. It can contain further other div elements if required.

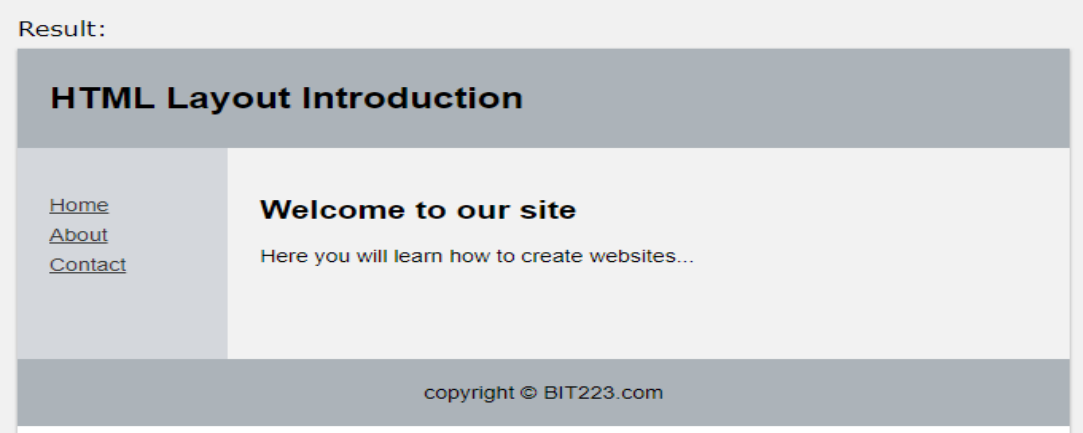
```
!DOCTYPE html>
head>
title>HTML Div Layout</title>
style>
    body {
        font: 14px Arial,sans-serif;
        margin: 0px;
    }
    .header {
        padding: 10px 20px;
        background: #acb3b9;
    }
    .header h1 {
        font-size: 24px;
    }
    .container {
        width: 100%;
        background: #f2f2f2;
    }
    .nav, .section {
        float: left;
        padding: 20px;
        min-height: 170px;
        box-sizing: border-box;
    }

    .nav {
        width: 20%;
        background: #d4d7dc;
    }
    .section {
        width: 80%;
    }
    .nav ul {
        list-style: none;
        line-height: 24px;
        padding: 0px;
    }
    .nav ul li a {
        color: #333;
    }
    .clearfix:after {
        content: ".";
        display: block;
        height: 0;
        clear: both;
        visibility: hidden;
    }
```

```
    .footer {
        background: #acb3b9;
        text-align: center;
        padding: 5px;
    }
</style>
</head>
<body>
    <div class="container">
        <div class="header">
            <h1>HTML Layout Introduction</h1>
        </div>
        <div class="wrapper clearfix">
            <div class="nav">
                <ul>
                    <li><a href="#">Home</a></li>
                    <li><a href="#">About</a></li>
                    <li><a href="#">Contact</a></li>
                </ul>
            </div>
            <div class="section">
                <h2>Welcome to our site</h2>
                <p>Here you will learn how to create websites...</p>
            </div>
        </div>

         <div class="footer">
             <p>copyright &copy; BIT223.com</p>
         </div>
     </div>
</body>
</html>
```

Result:




## Technical code terms explanations

. header {  point to **note** value **px** and unit**10** i.e  px10

Should not be separated.

padding: 10px 20px; {10px **represents height** 20px width}

background: #acb3b9;

 **container is the overall layout defined for each screen size HTML elements like: <article><section><Form> and more.in short box layout that includes from opening tag to closing tag.**

**We have two container classes**

**.container class –** provides a responsive fixed width container.

**.container-fluid class –**provides full width of the view point.

 term wrapper we give it a class, and that class is responsible for encapsulating (bidding)**all visual elements** on the page.

Clearfix – is a way for an element to **clear** its child automatically without any additional markup.

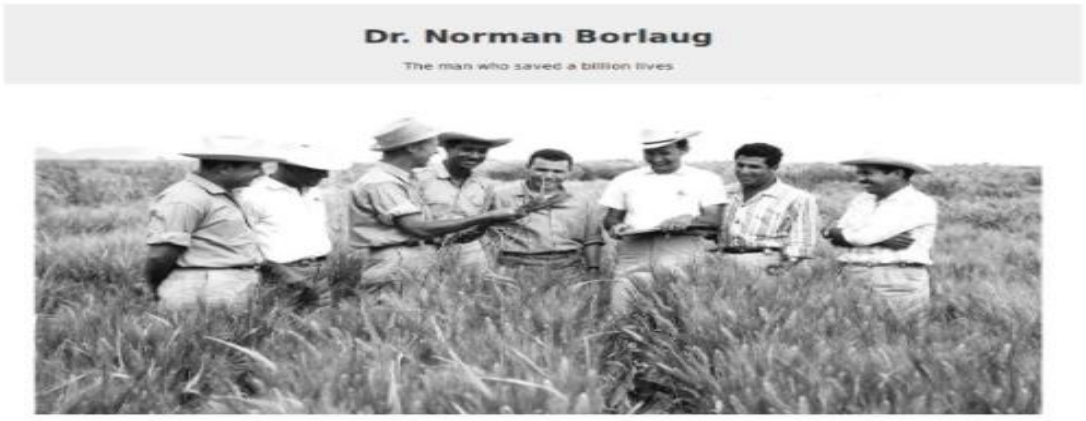
Definition 2: - Quickly and easily clear floated content within a container by adding clear fix utility.

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# Assignment one practical Question

**Due to be submitted on week 7 total marks (10 marks)**

**1. A TRIBUTE PAGE**




Write a tribute of someone you admire and publish as a webpage. This project will involve working with adding image, nav, links, lists and paragraphs. This project will require knowledge of HTML to create. However, you can use a bit of CSS to make the project look better. Use HTML page layout format.